

# Variable Length Subnet Mask

Course Code: CSC 3116

Course Title: Computer Networks



**Dept. of Computer Science**  
**Faculty of Science and Technology**

|                             |                                                |                 |  |                  |            |
|-----------------------------|------------------------------------------------|-----------------|--|------------------|------------|
| <b>Lecturer No:</b>         | 3                                              | <b>Week No:</b> |  | <b>Semester:</b> | Fall 25-26 |
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# Lecture Outline



1. Introduction
2. Subnetting
3. VLSM
  - Steps of VLSM
  - Example 1
  - Example 2
4. Homework

# Introduction



## Problems of Large Network

- A large network is difficult to manage
- Less security as each host of the network can reach all other hosts.
- Huge broadcast domain, thereby large bandwidth consumption
- Waste of unused IP addresses

# Introduction....



## How to overcome the problems?

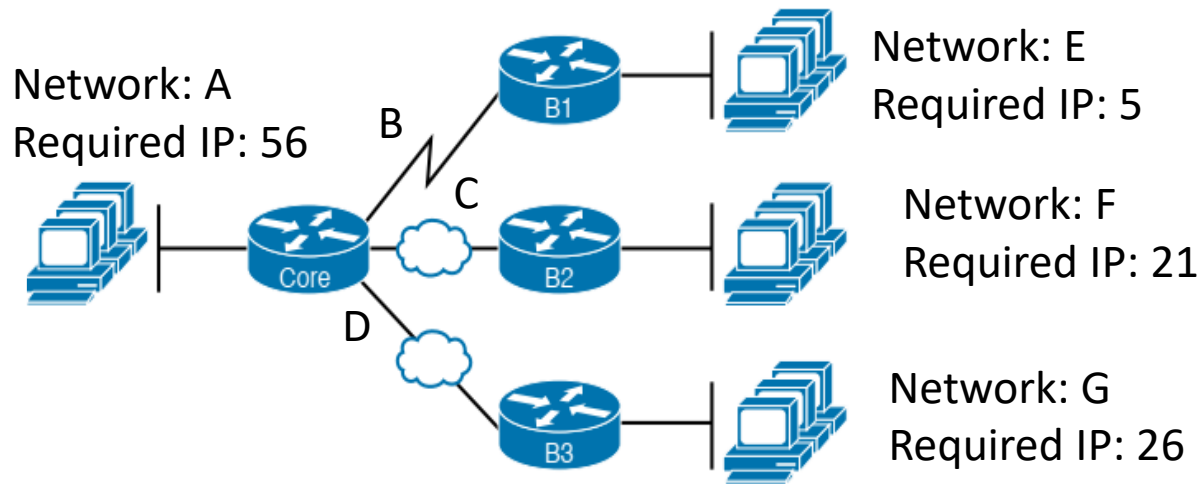
Dividing a large network into several smaller networks, called subdivided networks or subnets.

**Subnetting** is the strategy used to partition a single physical network into more than one smaller logical sub-networks (subnets). An IP address includes a network segment and a host segment. Subnets are designed by accepting bits from the IP address's host part and using these bits to assign a number of smaller sub-networks inside the original network. Subnetting allows an organization to add sub-networks without the need to acquire a new network number via the Internet service provider (ISP). Subnetting helps to reduce the network traffic and conceals network complexity. Subnetting is essential when a single network number has to be allocated over numerous segments of a local area network (LAN).

# Introduction....



For classful addressing



Networks: B, C, D  
Required IP: 4 (each)

| Network | No. of IP addresses allocated |
|---------|-------------------------------|
| A       | 256                           |
| B       | 256                           |
| C       | 256                           |
| D       | 256                           |
| E       | 256                           |
| F       | 256                           |
| G       | 256                           |

# Introduction



- Required IPs:  $56+5+21+26+4+4+4=120$
- **Classful address**
  - No. of allocated IPs:  $256 \times 7 = 1792$  IP
  - Percentage of unused IP =  $\frac{(1792-120) \times 100}{1792} = 93\%$  (approx.)

# Subnetting



**Subnetting** is done by taking the **bit/s from host part and adding it to the network part**. Remember, the first three octets of a Class C network is used to represent the network and the last octet is used to represent the host. The default format for a Class C IPv4 address is N.N.N.H.

To make things easy, you may remember this.

- ☐ If all the bits in the host part are "0", that represents the network id.
- ☐ If all the bits in the host part are "0" except the last bit, it is the first usable IPv4 address.
- ☐ If all the bits in the host part are "1" except the last bit, it is the last usable IPv4 address.
- ☐ If all the bits in the host part are "1", that represents the directed broadcast address.

All the IPv4 addresses between the first and last IPv4 addresses (including the first and last) can be used to configure the devices.

# Subnetting (Class -C)

## Example



**1 bit subnetting:** [ $2^1 \text{ bit} = 2 \text{ networks}$ ,  $2^7 \text{ bits} - 2 = 126 \text{ hosts}$ ]

If we include one bit from the host part to the network part, the subnet mask is changed into 255.255.255.128. The single bit can have two values in last octet, either 0 or 1.

**192. 168.10.0/24**

11000000.10101000.00001010.0 | 0000000 [Given IP Address]

11111111 .11111111 . 11111111.1 | 0000000 [Subnet Mask]

| SN No | Description        | Binaries                                    | Decimal               |
|-------|--------------------|---------------------------------------------|-----------------------|
| 1     | Network Address    | 11000000.10101000.00001010. <b>00000000</b> | <b>192.168.10.0</b>   |
|       | First IPv4 address | 11000000.10101000.00001010. <b>00000001</b> | 192.168.10.1          |
|       | Last IPv4 address  | 11000000.10101000.00001010. <b>01111110</b> | 192.168.10.126        |
|       | Broadcast Address  | 11000000.10101000.00001010. <b>01111111</b> | 192.168.10.127        |
| 2     | Network Address    | 11000000.10101000.00001010.10000000         | <b>192.168.10.128</b> |
|       | First IPv4 address | 11000000.10101000.00001010.10000001         | 192.168.10.129        |
|       | Last IPv4 address  | 11000000.10101000.00001010.11111110         | 192.168.10.254        |
|       | Broadcast Address  | 11000000.10101000.00001010.11111111         | 192.168.10.255        |



# Subnetting (Class -C)

## Example



**2 bits subnetting:** [ $2^2 \text{ bit} = 4 \text{ networks}$ ,  $2^6 \text{ bits} - 2 = 62 \text{ hosts}$ ]

The two bits added to network part can have four possible values in last octet and that are 00, 01, 10 and 11. That means, we can get four networks if we do a two bit subnetting.

11000000.10101000.00001010.00 | 000000

11111111 .11111111 .11111111 .11 | 000000

| SN No | Description        | Binaries                                     | Decimal        |
|-------|--------------------|----------------------------------------------|----------------|
| 1     | Network Address    | 11000000.10101000.00001010. <b>00</b> 000000 | 192.168.10.0   |
|       | First IPv4 address | 11000000.10101000.00001010. <b>00</b> 000001 | 192.168.10.1   |
|       | Last IPv4 address  | 11000000.10101000.00001010. <b>00</b> 111110 | 192.168.10.62  |
|       | Broadcast Address  | 11000000.10101000.00001010. <b>00</b> 111111 | 192.168.10.63  |
| 2     | Network Address    | 11000000.10101000.00001010.01000000          | 192.168.10.64  |
|       | First IPv4 address | 11000000.10101000.00001010.01000001          | 192.168.10.65  |
|       | Last IPv4 address  | 11000000.10101000.00001010.01111110          | 192.168.10.126 |
|       | Broadcast Address  | 11000000.10101000.00001010.01111111          | 192.168.10.127 |

# Subnetting (Class -C)

## Example



|   |                    |                                     |                |
|---|--------------------|-------------------------------------|----------------|
| 3 | Network Address    | 11000000.10101000.00001010.10000000 | 192.168.10.128 |
|   | First IPv4 address | 11000000.10101000.00001010.10000001 | 192.168.10.129 |
|   | Last IPv4 address  | 11000000.10101000.00001010.10111110 | 192.168.10.190 |
|   | Broadcast Address  | 11000000.10101000.00001010.10111111 | 192.168.10.191 |
| 4 | Network Address    | 11000000.10101000.00001010.11000000 | 192.168.10.192 |
|   | First IPv4 address | 11000000.10101000.00001010.11000001 | 192.168.10.193 |
|   | Last IPv4 address  | 11000000.10101000.00001010.11111110 | 192.168.10.254 |
|   | Broadcast Address  | 11000000.10101000.00001010.11111111 | 192.168.10.255 |

# Subnetting (Class -C)

Example



**Subnet Mask: 255.255.255.192**

- **How many subnet bits are used in this mask?**
- **How many host bits are available per subnet?**
- **What are the subnet addresses?**
- **What is the broadcast address of each subnet?**
- **What is the valid host range of each subnet?**

# Subnetting (Class -B)

## Example



**1 bit subnetting: [ $2^1 \text{ bit} = 2 \text{ networks}$ ,  $2^{15} \text{ bits} - 2 = 32766 \text{ hosts}$ ]**

If we include one bit from the host part to the network part, the subnet mask is changed into 255.255.128.0 The single bit can have two values in third octet, either 0 or 1.

10101100.00010000.0 | 00000000.00000000      [172.16.0.0]  
11111111 .11111111 .1 | 00000000.00000000

| SN No | Description        | Binaries                            | Decimal        |
|-------|--------------------|-------------------------------------|----------------|
| 1     | Network Address    | 10101100.00010000.00000000.00000000 | 172.16.0.0     |
|       | First IPv4 address | 10101100.00010000.00000000.00000001 | 172.16.0.1     |
|       | Last IPv4 address  | 10101100.00010000.01111111.11111110 | 172.16.127.254 |
|       | Broadcast Address  | 10101100.00010000.01111111.11111111 | 172.16.127.255 |
| 2     | Network Address    | 10101100.00010000.10000000.00000000 | 172.16.128.0   |
|       | First IPv4 address | 10101100.00010000.10000000.00000001 | 172.16.128.1   |
|       | Last IPv4 address  | 10101100.00010000.11111111.11111110 | 172.16.255.254 |
|       | Broadcast Address  | 10101100.00010000.11111111.11111111 | 172.16.255.255 |

# Subnetting



Is there any way to further reduce  
the unused IP addresses?

**Yup!**

Instead of giving the same number of IP addresses to all subnets,  
allocate different number of IP addresses depending  
on each subnet's needs.

# VLSM



- Variable Length Subnet Mask (VLSM)
- Instead of allocating the same number of IP addresses to all networks, the number of IP addresses allocated to network depends on the network's need.
- Different network is provided with different number of IPs

VLSM is similar in concept and intent to Classless Inter Domain Routing (CIDR), which allows a single Internet domain to have an address space that does not fit into traditional address classes. VLSM was originally defined in IETF RFC 1812.

# VLSM



## **Key VLSM features include:**

- Simple network configuration
- Network IP addressing through empty subnet filling
- Greater efficiency than fixed-length subnet masks (FLSM)
- Streamlined routing, where a router functions only with a VLSM sequence, versus a full IP address

## **VLSMs are used by a variety of network systems and services, including:**

Intermediate System to Intermediate System Protocol (IS-IS)

Border Gateway Protocol (BGP)

Enhanced Interior Gateway Routing Protocol (EIGRP)

Cisco's Open Shortest Path First (OSPF)

# VLSM

## Steps of VLSM



- *Step 1:* Write the networks' names and IP requirements in **descending order** of IP requirements. A (56), G(26), F(21), E(5), B(4), C(4), D(4)
- *Step 2:* Find the **number of host bits** needed to satisfy the IP requirement of each subnet?
- *Step 3:* Based on the number of host bits, **find the subnet mask for each subnet.**
- *Step 4:* Allocate IPs to each subnet starting from the beginning of the IP block. Allocate IPs to subnet sequentially according to the sorted network sequence found in Step 1.



# VLSM

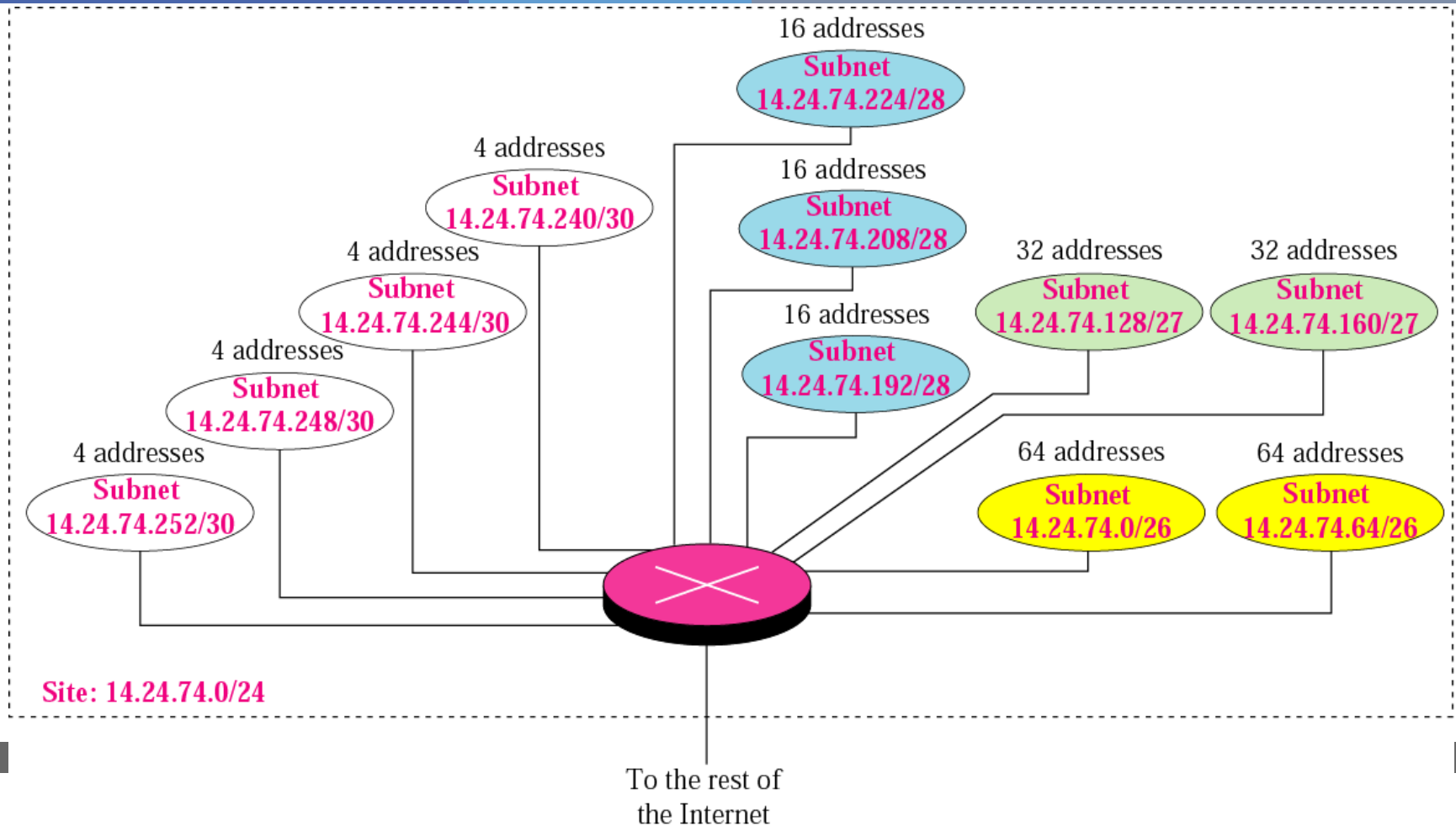


An organization is granted a block of addresses with the beginning address 14.24.74.0/24. There are a total of 256 addresses in this block. The organization needs the following **subnets** as shown below:

- a. Two subnets, each with 64 addresses.
- b. Two subnets, each with 32 addresses.
- c. Three subnets, each with 16 addresses.
- d. Four subnets, each with 4 addresses.

Design the subnets.

# VLSM



# VLSM....

## Example 1



11111111. 11111111. 11111111. 11000000

26

6

| Subnet | No. of IPs required | How many bits to borrow | No. of allocated IPs | No. of host bits<br>No. of net bits | Subnet mask     | Allocated IP range                 |
|--------|---------------------|-------------------------|----------------------|-------------------------------------|-----------------|------------------------------------|
| A      | 56                  | $2^6 > 56 > 2^5$        | 64                   | $x = 6,$<br>$y = 32 - 6 = 26$       | 255.255.255.192 | 192.168.5.0-<br>192.168.5.63/26    |
| G      | 26                  | $2^5 > 26 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 192.168.5.64-<br>192.168.5.95/27   |
| F      | 21                  | $2^5 > 21 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 192.168.5.96-<br>192.168.5.127/27  |
| E      | 5                   | $2^3 > 5 > 2^2$         | 8                    | $x = 3,$<br>$y = 32 - 3 = 29$       | 255.255.255.248 | 192.168.5.128-<br>192.168.5.135/29 |
| B      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.136-<br>192.168.5.139/30 |
| C      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.140-<br>192.168.5.143/30 |
| D      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 192.168.5.144-<br>192.168.5.147/30 |

# VLSM.....

## Example 1



## VLSM

No. of allocated IPs:  $64 + 32 + 32 + 8 + 4 + 4 + 4 = 148$  IP

Percentage of unused IP =  $\frac{(148 - 120) \times 100}{148} = 19\%$  (approx.)

Percentage of unused IP address (for the given network)

| Classful Addressing | VLSM |
|---------------------|------|
| 93%                 | 19%  |

# VLSM.....



Is it possible to further reduce the unused IP?

Spend your time to propose a new subnetting technique.

# VLSM....

## Example 2



- ❖ Suppose that we have three networks: A, B and C with IP requirements 50, 4 and 28. If You are given an IP block 130.3.0.0, allocate IPs performing subnetting.

| Subnet | No. of IPs required | How many bits to borrow | No. of allocated IPs | No. of host bits<br>No. of net bits | Subnet mask     | Allocated IP range           |
|--------|---------------------|-------------------------|----------------------|-------------------------------------|-----------------|------------------------------|
| A      | 50                  | $2^6 > 50 > 2^5$        | 64                   | $x = 6,$<br>$y = 32 - 6 = 26$       | 255.255.255.192 | 130.3.0.0-<br>130.3.0.63/26  |
| C      | 28                  | $2^5 > 28 > 2^4$        | 32                   | $x = 5,$<br>$y = 32 - 5 = 27$       | 255.255.255.224 | 130.3.0.64-<br>130.3.0.95/27 |
| B      | 4                   | $2^2 = 4$               | 4                    | $x = 2,$<br>$y = 32 - 2 = 30$       | 255.255.255.252 | 130.3.0.96-<br>130.3.0.99/30 |

# Lab Task



1. Suppose that we have three networks: A, B and C with IP requirements 100, 300 and 2008, respectively. If You are given an IP block 130.3.0.0, allocate IPs performing subnetting.
2. Suppose that we have six networks: A, B, C, D, E and F with IP requirements 120, 400, 9, 40, 32, and 7, respectively. If You are given an IP block 210.3.0.0, allocate IPs performing subnetting.
3. Consider your ID: AB-CDEFG-H. Now, suppose that we have four networks: P, Q, R, and S with IP requirements  $D*100$ ,  $E*100$ ,  $F*100$ , and  $G*100$ , respectively. If You are given an IP block 172.16.0.0/16, allocate IPs performing subnetting.



# References

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.





## Recommended Books

1. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
2. **CCNA Routing and Switching**, *T. Lammle*, John Wily & Sons, Second Edition, 2016, USA.